Phase I – Business Process Explanation

Process: Daily Fare Collection and Revenue Reconciliation | BusFareTrack | Uwimana Aline (20251SEN260)

1. The business process

This process follows a bus fare from the moment a passenger taps a smartcard to the moment the day's revenue is checked against what was expected. Its purpose is to make sure that every fare paid is recorded once and correctly, and that missing or unusually low revenue is noticed on the same day. It addresses revenue leakage, which happens when fare records are scattered or checked by hand (The New Times, 2025). It matters to the organization because fare revenue pays for fuel, maintenance and salaries.

2. Actors and responsibilities

Passenger: pays the fare by tapping a smartcard. **Smartcard validator (with the driver):** checks the card and sends the fare to the database. **BusFareTrack database:** validates and stores each fare, totals the revenue and raises alerts. **Revenue officer:** starts the end-of-day closing. **Operations manager:** reviews alerts and decides what action to take.

3. How the process runs

The process starts when a passenger boards and taps the card. The validator checks the card; if it is invalid or has too little credit, the fare is rejected and the process ends for that passenger. If it is valid, the fare goes to the database, which checks that the bus is in service and the trip is open. If not, the attempt is refused and written to the audit log. Otherwise the fare is saved. This repeats for every passenger during the day. At the end of the day the revenue officer closes the day. The database totals the fares per route and vehicle and compares them with the expected revenue. If revenue is below the expected level, an alert is created and the manager reviews it, investigates (for example a faulty validator or missing taps), records the outcome and closes the alert. If revenue is normal, the day is marked reconciled and the process ends.

4. Information flow and data activities

Information moves from the passenger's card to the validator, then to the database, and finally to the revenue officer and manager as daily totals and alerts. The table lists the activities that touch data.

Activity	Data operation	Information generated or used
Check vehicle and trip	Retrieve	Vehicle status and open trip (VEHICLES, TRIPS)
Record fare	Create	Card hash, time, trip and amount (FARE_TRANSACTIONS)
Log rejected attempt	Create	User, time, reason (AUDIT_LOG)
Total day's revenue	Retrieve, Create	Reads fares; writes totals per route, vehicle and day (DAILY_REVENUE)
Close day	Update	Marks the day closed so its fares cannot change
Raise alert	Create	Route, vehicle, expected and actual revenue (REVENUE_ALERTS)
Review alert	Retrieve	Alert and daily revenue
Record outcome	Update	Alert status and manager's note
Archive	Archive	Fares are never deleted; old closed days may be archived

5. Support for management decisions

Daily totals and alerts let the manager compare routes and buses, find problems the same day and decide where to act, instead of finding losses weeks later.

Reference: The New Times. (2025, November 30). *How Kigali's new public transport model will work*. https://www.newtimes.co.rw/article/31605/news/infrastructure/how-kigalis-new-public-transport-model-will-work